CONSTRUCTION TECHNOLOGY / ENTERPRISE SAAS

PCM Tools

Connecting the field. Engineering control.

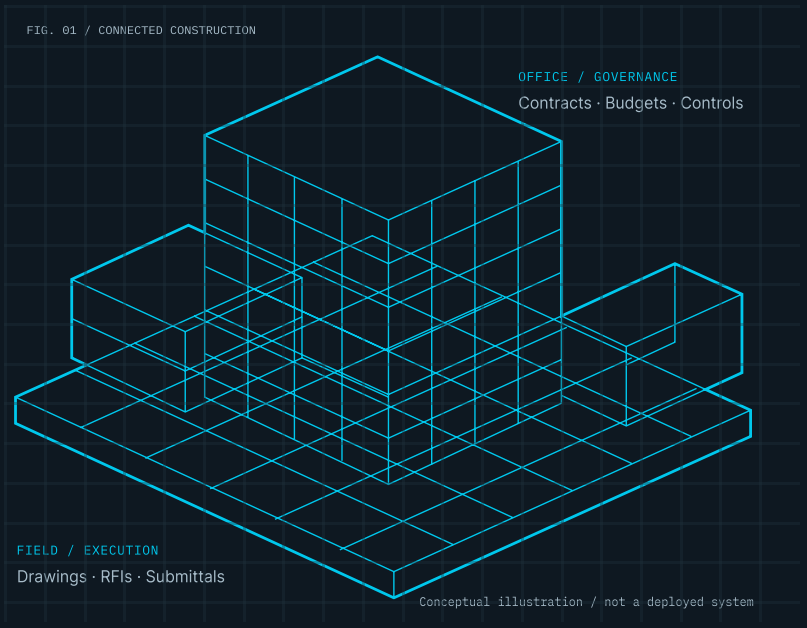
Enterprise Construction Management Software — a cloud ERP ecosystem connecting project operations, document control and financial management.

CLIENT / AAA PARTNERSHIPS / AAA PROJECTS

ENGINEERING PARTNER / TIME GLOBAL TECH

ENGAGEMENT / MAY 2019 – MAY 2021

FIG. 01 / CONNECTED CONSTRUCTION



42
MODULES ENGINEERED

2,016+
DEVELOPMENT HOURS

Web + Apps
MULTI-PLATFORM DELIVERY

SOURCE-REPORTED PROJECT SCOPE · ENGINEERING PORTFOLIO

One project. Many moving parts.

The challenge was not a single task. It was continuity across commercial, technical and site-level decisions.

Office executives
Visibility / planning / decisions

Project stakeholders
Bids / contracts / commitments

PCM Tools / Cloud ERP
A shared operational context

Field personnel
Drawings / RFIs / submittals

Finance & control
Budgets / ledgers / approvals

Conceptual relationship map.
Not an observed research journey.

THE FRICTION

Silos break continuity

Bidding, contractual commitments, field drawings and financial controls involve different parties. Disconnected records make it harder to keep decisions in context.

THE RESPONSE

Centralize the workflow

A cloud-based ERP ecosystem was conceptualized to centralize and automate these workflows, linking office oversight with the information generated on site.

THE TIME GLOBAL TECH REMIT

Full-lifecycle engineering

Time Global Tech was commissioned to build the backend architecture, enterprise web administration and native mobile applications as a connected delivery ecosystem.

01 **Structure the core**
Build modular application services for construction operations and enterprise financial workflows.

02 **Connect the interfaces**
Expose shared services to the web ERP, administration tools and native mobile clients.

03 **Support controlled delivery**
Apply review, testing and integration practices to a multi-platform engineering programme.

A modular ecosystem, not a feature list.

The source names eight engineering focus areas. The clusters below organize those areas without inventing a module inventory.

42

CUSTOM MODULES ENGINEERED



One marker per engineered module. The graphic does not identify modules or assign them to clusters.

CONFIGURABLE PLATFORM RANGE

26–55

Interconnected operational modules in the source's broader configuration range.

A	**Project & commercial operations** Construction Project Management Bidding and Estimating	Establish the project context and the commercial work that precedes contractual commitment.
B	**Technical & document control** Contract Management & RFIs Drawings & Document Submittals	Keep contractual questions and technical documentation connected to project execution.
C	**Financial management** Budgeting & Financial Commitments Enterprise Financial Management	Relate project commitments to the wider financial controls of the enterprise.
D	**Governance & platform scale** Administrative Security & Controls Multi-tier Cloud SaaS Architecture	Provide administrative oversight and a modular foundation across the delivery ecosystem.

SCOPE CLARIFICATION

42 is the engineered delivery count; 26–55 is the stated configurable platform range. These are different claims. The source does not list all 42 modules or establish that 55 were delivered.

Shared services. Multiple points of access.

A conceptual logical architecture based on the source description — not a documented deployment or infrastructure inventory.

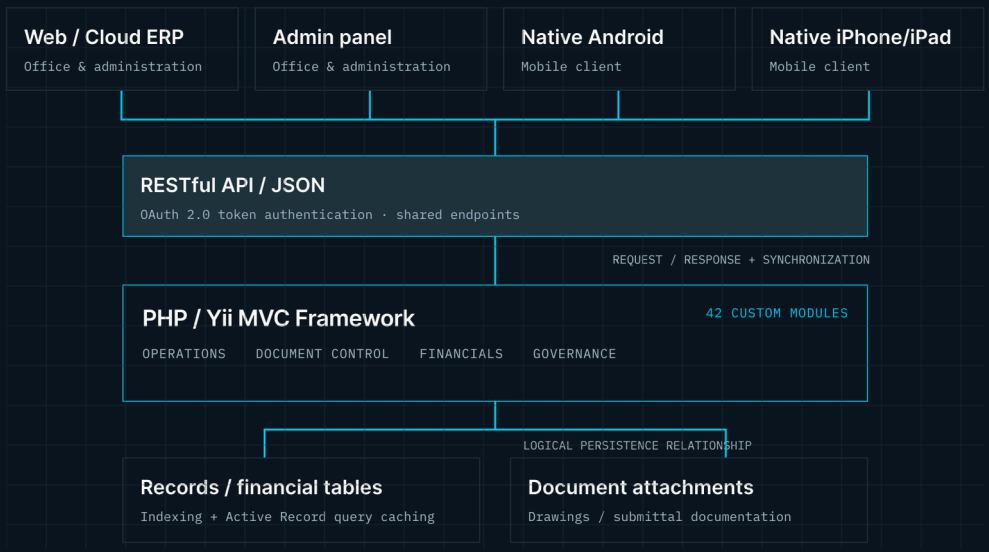


FIG. 02 / Lines indicate logical dependencies and data exchange, not physical network routes. No cloud vendor, database engine or tenancy-isolation implementation is specified in the source.

DESIGN CONSIDERATIONS

01 / Heavy site writes

Optimized Yii MVC and a modular service-oriented approach supported the source-described concurrent database writes from construction sites.

02 / Financial read paths

Indexing and Yii Active Record query caching targeted financial ledger and change-order queries. The reported latency result is scoped on page 08.

03 / Connected clients

REST/JSON endpoints linked dashboards and native iOS/Android apps for source-described real-time synchronization. OAuth 2.0 handles token authentication, not every security control.

Carry context from bid to budget.

An illustrative sequence across the named PCM Tools workflows. It is not a verified process map or a record of approval states.

01 Bidding

COMMERCIAL

Scope, estimate and bid information establish the commercial context for a project.

02 Contracts

COMMITMENT

Contractual commitments translate the commercial context into controlled project obligations.

03 RFIs & drawings

OFFICE ⇔ FIELD

Questions and technical drawings connect site conditions with office-level clarification.

04 Submittals

DOCUMENT CONTROL

Project documentation supplies the technical context needed for review and coordination.

05 Financial commitments

FINANCE

Budgets and commitments bring the financial consequences back into enterprise oversight.

OFFICE → FIELD

Send usable context

Recommended practice: attach the relevant project, contract and document references when a technical question moves to site personnel.

FIELD → OFFICE

Return the decision trail

Recommended practice: connect site updates to the originating RFI, drawing or submittal so the office can understand what changed.

OPERATIONS → FINANCE

Preserve the relationship

Recommended practice: relate contractual or technical changes to the relevant budget and commitment records before financial review.

> HOW TO READ THIS SEQUENCE
> Real construction workflows can branch, overlap or loop back. The sequence explains relationships; the handoff guidance is editorial advice, not a claim about historical implementation.

Different roles. A connected foundation.

Four delivery surfaces extended the ERP ecosystem from enterprise administration to construction-site activity.



WEB / CLOUD ERP

Operational visibility

Construction operations visibility, resource scheduling and financial accounting in the web-based cloud ERP.

ADMIN CONTROL PANEL

Enterprise administration

Tenant management, project-level RBAC, audit oversight and module configuration for administrative teams.

NATIVE ANDROID

Site management

Site managers and engineers log inspections, view drawings and update submittals from the native Android app.

NATIVE IPHONE / IPAD

Project oversight

Project timelines, budget metrics and direct client communications through the native iPhone/iPad applications.

FIG. 03 / Device outlines contain explanatory task labels, not production UI. Shared services express the logical relationship between platforms; no offline functionality is asserted.

Quality as a repeatable engineering loop.

Agile/Scrum delivery combined structured iteration with code review, automated tests and continuous integration.

CODE REVIEW

Review the change

Rigorous code reviews and PSR-12 PHP standards support readable, consistent implementation.

UNIT TESTS

Exercise core behavior

Automated unit testing pipelines check core-module behavior before integration.

STATIC ANALYSIS

Surface code risks

Static analysis and periodic vulnerability scans address OWASP Top 10 risk areas.

CONTINUOUS INTEGRATION

Integrate continuously

CI connects the engineering checks to the recurring development and integration workflow.

24

STRUCTURED DEVELOPMENT SPRINTS

Source-reported delivery count within the broader engineering engagement.

42

STORY POINTS / TWO-WEEK ITERATION

Reported average velocity. Story points are a planning measure, not a cross-team productivity benchmark.

88.4%

AUTOMATED TEST COVERAGE / CORE MODULES

Reported core-module coverage, not a claim of complete defect prevention.

CADENCE ≠ ENGAGEMENT DURATION

The engagement ran May 2019 – May 2021. The source reports 24 sprints and a two-week iteration cadence; it does not establish continuous sprint activity across the entire two-year period. All figures are source-reported.

Three figures. Three different meanings.

A query-level improvement, an API average and an availability commitment should never be presented as the same kind of evidence.

42%

REPORTED QUERY LATENCY REDUCTION

Scoped to financial ledger and change-order tables. The source attributes the reduction to database indexing and Yii Active Record query caching.



Normalized illustration derived from the reported reduction; not raw before/after timings. Source-reported; benchmark evidence not supplied.

REPORTED RESPONSE TIME

<185 ms

Average API response

Source-reported average. Endpoint mix, workload and test window are not provided; no percentile or load-capacity claim is implied.

SERVICE-LEVEL COMMITMENT

99.9%

Uptime SLA — not measured uptime

A stated availability commitment. The source does not provide monitoring history demonstrating achieved uptime.

BEFORE PUBLICATION / RECOMMENDED EVIDENCE

Confirm the test window

Record dates, software version and the environment used for each performance report.

Describe the workload

Identify endpoint and query mix, concurrency, dataset scale and caching conditions.

Approve the report

Check baseline comparability, averaging method and SLA terms against signed-off evidence.

Control is layered, not a single feature.

The source reports the following security and governance controls. This is an explanatory model, not an independent security assessment.



FIG. 04 / Conceptual control layers. Their placement explains complementary purposes, not network topology or a certified assurance boundary.

01 **OAuth 2.0 + project-level role-based permissions**
Token authentication identifies requests; project roles limit permitted actions within the project context.

02 **TLS 1.3 in transit**
Protects data moving between connected interfaces and services against interception in transit.

03 **AES-256 at rest**
Reported for database records and document attachments to protect stored project information.

04 **Immutable decision logs**
Records change orders, budget changes and submittal approvals to preserve a traceable decision history.

05 **Encrypted backups + point-in-time recovery**
Daily encrypted backups and point-in-time recovery support restoration; no restore-test results are supplied.

06 **Vulnerability scans + static analysis**
Periodic scans and CI/CD-integrated static analysis address OWASP Top 10 risks; they do not guarantee security.

PRIVACY / REPORTED ALIGNMENT, NOT CERTIFICATION

The source describes personnel-data anonymization for PIPEDA and claims compliance. Here this is presented as reported privacy alignment, not a certification or an independently verified legal-compliance conclusion.

From field execution to corporate governance.

A substantial engineering scope across web and native mobile — with a clear evidence trail still needed for external publication.

SOURCE-REPORTED DELIVERABLES

42
Custom modules engineered

2,016+
Logged development hours

Web + Apps
Cloud ERP, admin, Android, iPhone/iPad

The project linked construction operations, document workflows and enterprise financial governance. The source does not report measured customer productivity or revenue impact.

RECOMMENDED HANDOVER / PUBLICATION CHECKS — NOT DELIVERED WORK

01 **Module inventory & configuration**
Reconcile the 42 engineered modules with an approved inventory; explain the separate 26–55 configurable range.

02 **Performance & SLA evidence**
Confirm query and API benchmark reports, test windows and workloads; retain the agreed uptime SLA terms.

03 **Coverage & quality evidence**
Archive the core-module coverage report, its scope and build version, with review and CI records.

04 **Recovery evidence**
Review encrypted-backup records, point-in-time recovery capability and documented restoration-test evidence.

05 **Access matrix & privacy review**
Validate project roles and access rights; obtain an appropriate privacy review of personnel-data handling and PIPEDA alignment.

06 **Client naming & publication approval**
Confirm approved use of AAA PARTNERSHIPS / AAA PROJECTS, project dates, technical claims and portfolio wording.

Connected construction means carrying project context between the field and the office.

Source: client-supplied PCM Tools Portfolio Case Study; reported metrics and controls have not been independently verified.